

not have any security features to prevent widespread piracy of licensed music.

A completely free alternative

Let's see what else you can really depend on for storing and playing your collection of songs. Ogg Vorbis could be just the thing for you in case you want quality and, more importantly, you want it free for distribution. Ogg Vorbis is open source code, patent free and you don't have to pay a penny for giving your own songs to anyone. As of now, it is the only audio format that does this.

Ogg Vorbis uses different mathematical principles from MP3 in how it encodes its music and plays it. The Ogg Vorbis Codec works on the basis of VBR like MP3 but does it more efficiently. In the standard MP3 format, VBR allows for a rise and decline of 32 kbps steps. In Ogg Vorbis, the jump is of 1 kbps, which means you get very accurate quality and file sizes are strictly kept to a minimum. Though Ogg Vorbis files themselves can be encoded between 8 kbps to 512 kbps, the current beta encoder supports 128, 160, 192, 256 and 350 kbps in mono or stereo.

Since streaming audio is a very important application for these file formats in the future, any successful format would need to include support for them. This is natively supported within the Ogg Vorbis Codec. Currently, the designers of Ogg Vorbis are working with the makers of Icecast, an open source streaming server which will allow users to enjoy streaming audio.

Similar to what MP3's ID3 tag offers, Ogg Vorbis supports a proprietary comment header system, which allows you to include various information about the track such as title, artist, song name, and so on.

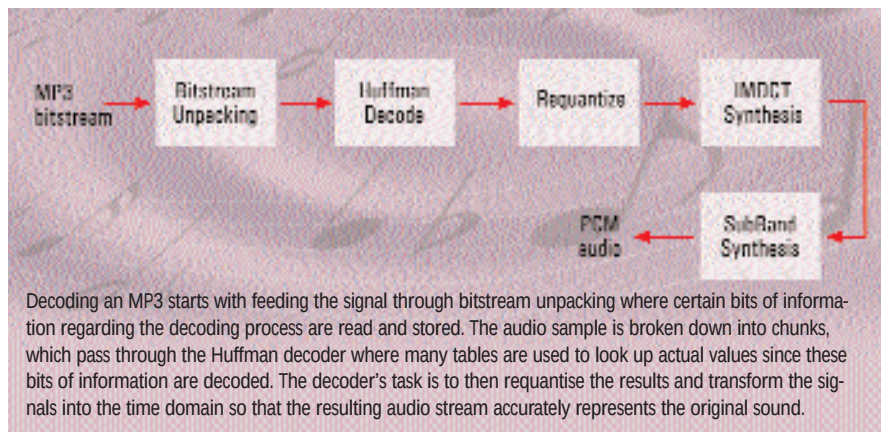
Ogg Vorbis is patent free, open source and thus can be distributed and used for personal and commercial purposes. Similarly, codecs are being developed for other operating systems like Linux, BeOS, etc. Ogg Vorbis files can be played back using Winamp and other supportive audio players simply by using an Ogg Vorbis plug-in.

How can we forget Microsoft?

Microsoft likes doing things its own way. After having taken over the operating system and office suite market, they've set sights upon pushing their audio formats in a similar fashion with their Windows Media Audio.

What you are about to get is Windows Media Audio 8—Microsoft's latest iteration of the WMA standard and their idea of the perfect competition to MP3Pro and Ogg Vorbis.

The format is similar to MP3Pro in terms of size of the file. The bandwidth used is 48 kbps for near CD-quality and 64 kbps for CD-quality audio, similar to what MP3Pro offers. The format has been created for streaming of media



over 56 kbps dial-up connections to high-speed DSL and cable modem connections.

One particular feature incorporated into WMA 8 that should appeal to the record industry is a technology called Digital Rights Management, or DRM for short. DRM is an end-to-end solution for encrypting and distributing digital content in a secure manner on the Internet.

The Windows Media format is a compilation of various components, which is called as a Windows Media File Container. It contains WMA and WMV using the codecs along with DRM technology. Metadata is similar to MP3's ID3 tag as it contains information about the artist, song name, etc. The format also has well integrated voice recognition and text to speech capabilities using Microsoft speech engine.

What's your format?

So finally, which is the best format? If you're looking at availability, MP3 is no doubt the most happening audio format available. MP3Pro has just started and the demo encoder/player does not have support for encoding audio at full quality yet. For the time being, one will have to stick to encoding at 64 kbps using MP3Pro.

Ogg Vorbis offers high quality audio, thanks to the optimal VBR encoding and playing features, but availability and widespread industry support still remains a major problem. And without security and encryption, it is difficult for the record industry to adopt Ogg Vorbis as their audio standard du jour. The latest version of the Ogg Vorbis Codec, V1.0, will soon be available and proper encoders and players should also be available for the general public to adopt.

Microsoft is doing a splendid job with their WMA 8 standard. With high compression, integrated security features and streaming capability, WMA 8 appears to have the most potential to be the prominent audio standard in the future. At the moment, while WMA is catching ground fast with portable audio player makers' support in their devices, MP3 still remains the format of choice for most device manufacturers and users. 

What to look for in a Protected Player

With audio standards changing the way they are today, it would be best to scout for a player that will support a particular format in the future. To ensure that your digital audio player is indeed future-proof, purchase a unit equipped with upgradable firmware. Be sure you buy from a reputed company, as the inclusion of upgradable firmware does not guarantee that the company will provide you with an upgrade in the future.

Several memory-based and MP3/CD players are available that permit firmware updates. Son-icblue's Rio players (500 and above) support MP3, WMA and Audible formats. If an MP3 CD player suits your fancy, take a look at the AVC Soul Player which has a very good track record of providing constant firmware updates through their Web site.